

Escape Room Reflection

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For our resource management game, we created a balloon tower defense style game. It expands on the narrative using the characters from the dungeon crawler portion of the story as towers. I enjoy this type of resource management game because it keeps the gameplay engaging (there's a lot happening on the screen testing the player's ability to multitask), and strategy focused (which order to place and upgrade towers). The main resource players are managing will be currency needed to improve defenses and game knowledge of enemy spawn patterns. We worked on the same Unity game engine with the same style 3D world with 2D enemy Doom style mechanics as in the previous two games. The delivery timeline was more smooth than in our past games and we did a good job of scoping.

I think as a team we improved a lot at making games over the course of the semester. And this game came together rather well as a reflection of that. I was much more comfortable working in a three dimensional environment and with the unity game engine. It made it significantly easier to add features and tie in the features I made into the rest of the team's work. I worked on creating enemies, enemy paths, and the wave spawning mechanics. The enemies are the same enemies the player faced in the dungeon crawler, as it is a continuation of that narrative. Individual rounds within waves can have a set number of enemies, different enemy types, a speed, and a spacing between the enemies spawned. All these variables can change between rounds. The pathing was created using waypoints preset in the map. Enemies spawn at waypoints and then follow along in a specified ordering depending on the round.

Unfortunately, I got very sick over the week between the beta and the final submission. I was not able to add the multiple enemy types that I was planning on. Ideally they would have

been with special mechanics, building off the base enemy type with weaknesses to different types of towers. That would have added to the overall strategy as different towers would be better at defeating different types of waves.

I liked our map design and overall aesthetics. I think it flows well together with the narrative as well. Props from the escape room are scaled up and used off to the sides. The cast of characters from the dungeon crawler are back as the towers. The enemies are similar to the ones faced by the player and his team in the dungeon crawler as well. If we had more time it would have been great to add some meta progression system that could track player stats and perhaps have some special upgrades they could unlock like new tower types or new tower abilities. That would add to the replayability. I think it would also be an easy game to scale as all the core mechanics are present, just new map design would be needed. We also should have been a little more ambitious with the map. Once the key mechanics were created like the towers and the enemies, it would have been easy to put them into a much bigger map. I think the strategy component would scale with the map size as there's more decisions about where to place towers, which towers to upgrade, and which key choke points are important in the current round. It would also make it easier to scale difficulty, because if enemy speed is fast on a small map it can be kind of hard to follow visually. But with a larger map, enemies can be faster without making the visual experience so jumpy as enemies have larger distances to travel.

I liked the enemy waypointing, especially with the multiple paths and paths changing after each round. This increases the difficulty and replayability of the game by forcing the player to learn spawn patterns. It also adds depth to the strategy because some of the towers players place may not be used at all in some of the rounds or they may be critical to making it through that round.

I think the team did a much better job of scoping our work this time around. In previous projects we started off with very ambitious goals and when we failed to meet those high expectations, motivation fell off and it always felt like we were behind. However in this game, we were much more rigid with using a small scope for the beta, getting all the key mechanics working and polished, and then scaling up only once the core gameplay was functional. I think this was a much smarter approach because it enabled us to focus on what was important first and meet the expectations for our game. Then we could build on top of that motivation and add more aesthetic, difficulty, and other nice to have elements on top of an already functional project. Looking back I think that was a key problem in some of our past games. Not prioritizing a functional project first. Having the key components present by the beta made it much easier to scale up. It also helped with integration issues because trying to implement more complex mechanics and then integrate them all at once has caused problems for us in the past.

Overall I think this project was well done and as a team we improved over the course of the semester. Unfortunately, I was not able to implement all the mechanics I would have liked to due to illness. But the game plays well and integrates well into our overall narrative.